

Messrs. : 一般共用

Date : 2025/04/14

APPROVAL SHEET

Product Name : AUTOMOTIVE Soft Termination Multilayer Ceramic
Chip Capacitors

Part No. : MT Soft Termination Capacitors Series

Description : Size 0402 to 2220, X7R/X7S/C0G, 10Vdc to 3000Vdc

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SPECIFICATION

FOR

Product Name : AUTOMOTIVE Soft Termination Multilayer Ceramic
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SPEC. No. : MT-000-001-23

DATE : 2025/04/14

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1. INTRODUCTION

MLCC consists of a conducting material and electrodes. To manufacture a chip-type SMT and achieve miniaturization, high density and high efficiency, ceramic condensers are used.

PDC's MT Soft termination series MLCC is made by X7R/X7S/C0G(NPO) dielectrics and which provides product with high electrical precision, stability and reliability. Besides, MT Soft termination series MLCC is tighten controlling in quality in line to assure quality performance in automotive applications and qualified to AEC-Q200.

2. FEATURES

- A wide selection of sizes is available (0603 to 1812).
- High capacitance in given case size.
- Capacitor with lead-free termination (pure Tin).
- The MT Soft termination series meet AEC-Q200 requirement.
- RoHS Compliant.

3. APPLICATIONS

- For Navigation & Information equipments.
- For entertainment equipments.
- For comfortable equipments.
- For Automotive electronic equipment.

4. HOW TO ORDER

MT	31	X	471	K	101	E	E	E
PDC Family	Size	Dielectric	Capacitance	Tolerance	Rated Voltage	Packaging	Thickness	Control Code
Table 1	Table 2	Table 3	Table 4	Table 5	Table 6	Table 7	Table 8	Table 9

Table 1	PDC Family
Code	Description
MT	Automotive Capacitor Qualified to AEC-Q200

Table 2		Size			
Code	Description	Code	Description	Code	Description
18	0603(1608)	32	1210(3225)	55	2220 (5750)
21	0805(2012)	42	1808(4520)		
31	1206(3216)	43	1812(4532)		

Table 3	Dielectric Material Characteristics		
Code	Description	Code	Description
X	X7R	A	X7S
N	C0G(NPO)		

Table 4		Capacitance Rule Code	
Code	Description	Code	Description
R47	0.47pF	102	102=10x10 ² =1000pF
0R5	0.5pF	104	104=10x10 ⁴ =100nF
100	100=10x10 ⁰ =10pF	106	106=10x10 ⁶ =10μF

Table 5		Tolerance			
Code	Description	Code	Description	Code	Description
A	±0.05 pF	I	-10% ~ 0%	Q	±0.03 pF
B	±0.10 pF	J	±5 %	Z	-20% ~ +80%
C	±0.25 pF	K	±10 %	X	+10%~+20%
D	±0.50 pF	L	0% ~ +10%		
F	±1 %	M	±20 %		
G	±2 %	N	-5% ~ +10%		
H	±3 %	P	±0.02 pF		

Table 6		Rated Voltage			
Code	Description	Code	Description	Code	Description
100	10Vdc	101	100Vdc	102	1000Vdc
160	16Vdc	251	250Vdc	202	2000Vdc
250	25Vdc	501	500Vdc	302	3000Vdc
500	50Vdc	631	630Vdc		

Table 7		Packaging Type	
Code	Description	Code	Description
B	Bulk	T	Tray package
E	Tape and 7" Reel, Embossed Tape	P	Tape and 7" Reel, Paper Tape
K	Tape and 10" Reel, Embossed Tape	D	Tape and 10" Reel, Paper Tape
L	Tape and 13" Reel, Embossed Tape	G	Tape and 13" Reel, Paper Tape
F	Tape and 13" Reel, Embossed Tape Reduced packaging		

Table 8		Thickness Description			
Code	Description	Code	Description	Code	Description
A	0.60 ± 0.10 mm	I	1.25 ± 0.20 mm	Q	0.50+0.02/-0.05 mm
B	0.8 + 0.15/-0.10 mm	J	1.15 ± 0.15 mm	R	3.10 ± 0.30 mm
C	1.25 ± 0.10 mm	K	0.50 ± 0.20 mm	S	0.80 ± 0.07 mm
D	1.40 ± 0.15 mm	L	0.30 ± 0.03 mm	T	0.85 ± 0.10 mm
E	1.60 ± 0.20 mm	M	0.95 ± 0.10 mm	U	0.50 ± 0.10 mm
F	2.00 ± 0.20 mm	N	0.50 ± 0.05 mm	V	0.20 ± 0.02 mm
G	2.50 ± 0.30 mm	O	3.50 ± 0.20 mm	X	0.80 ± 0.10 mm
H	2.80 ± 0.30 mm	P	1.60 +0.3/-0.10 mm	Z	0.25 ± 0.03 mm

Table 9	Special Control Code
Code	Description
E	Soft Termination
Z	Anti-Arcing+Anti-Bendin



5. EXTERNAL DIMENSIONS

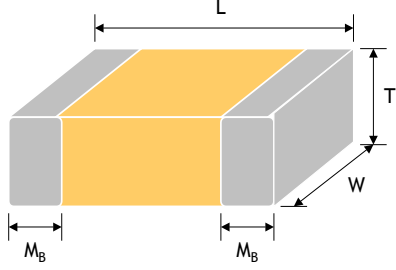
Size Inch (mm)	L (mm)	W (mm)	Code / T (mm)	M _B (mm)	
0402 (1005)	1.00±0.20	0.50±0.20	K	0.25 +0.05/-0.10	
0603(1608)	1.60±0.20	0.80±0.10	S	0.40±0.15	
	1.60±0.30	0.80±0.30	B		
0805(2012)	2.10±0.20	1.25±0.20	See No.4 Reference Table 8	0.50±0.20	
1206(3216)	3.30±0.30	1.60±0.20 +0.3/-0.1 [#] ▽1.60±0.40		0.60±0.20	
1210(3225)	3.30±0.40	2.50±0.30 ▽2.50+0.35/-0.30		0.75±0.35	
1812(4532)	4.60±0.50	3.20±0.30		0.75±0.35	
2220 (5750)	5.70±0.50	5.00±0.40		0.85±0.35	

Fig. 5.1 The outline of MLCC

[#] For 1206 P thickness products (P)

▽ For 1206/Cap.≥5.6μF & 1210/Cap.≥12μF products

6. GENERAL ELECTRICAL DATA

Dielectric	C0G(NPO)		X7R/X7S
Size	1206, 1210		,0402,0603, 0805, 1206, 1210, 1812,2220
Rated voltage (WVDC)	630V, 1000V		10V, 16V, 25V, 50V, 100V, 250V, 630V, 1000V, 2000V, 3000V
Capacitance range	1.5pF ~ 22nF		220pF ~ 22μF
Capacitance tolerance	Reference to Table 5		
Tan δ	Cap. Range	Q Spec.	Follow No.17 of 8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS
	Cap.<30pF	Q≥400+20C	
	Cap.≥30pF	Q≥1000	
Capacitance & Tan δ Test condition	Measured at the condition of 30~70% related humidity		
	For 25°C at ambient temperature		Preconditioning for Class II MLCC : Perform a heat treatment at 150±10°C for 1 hour, then leave in ambient condition for 24±2 hours before measurement
	Cap. Range	Test Condition	1.0±0.2Vrms, 1.0KHz±10%, at 25°C ambient temperature
	Cap.≤1000pF	1.0±0.2Vrms, 1.0MHZ±10%	
	Cap.>1000pF	1.0±0.2Vrms, 1.0KHZ±10%	
Insulation resistance at Ur	≥10GΩ or RxC≥500Ω-F, whichever is smaller		≥10GΩ or RxC≥100Ω-F, whichever is smaller
Operating temperature	-55°C to +125°C		
Capacitance characteristic	±30ppm/°C		X7R : ±15% X7S : ±22%
Termination	Cu/Ag polymer/Ni/Sn (lead-free termination)		



7. CAPACITANCE RANGE

7-1 C0G(NPO)

Cap(pF)	EIA Size Code	1206	1210	
		630V	630V	1000V
0.1	0R1			
0.2	0R2			
0.3	0R3			
0.4	0R4			
0.5	0R5			
1.0	1R0			
1.2	1R2			
1.5	1R5	X		
1.8	1R8	X		
2.2	2R2	X		
2.7	2R7	X		
3.3	3R3	X		
3.9	3R9	X		
4.7	4R7	X		
5.6	5R6	X		
6.8	6R8	X		
8.2	8R2	X		
10	100	X	M	M
12	120	X	M	M
15	150	X	M	M
18	180	X	M	M
22	220	X	M	M
27	270	X	M	M
33	330	X	M	M
39	390	X	M	M
47	470	X	M	M
56	560	X	M	M
68	680	X	M	M
82	820	X	M	M
100	101	X	M	M
120	121	X	M	M
150	151	X	M	M
180	181	X	M	M
220	221	X	M	M
270	271	X	M	M
330	331	X	M	M
390	391	X	M	M
470	471	X	M	M
560	561	X	M	C
680	681	X	M	C
820	821	X	M	C
1000	102	X	M	E/F
1200	122		M	E/F
1500	152		M	F
1800	182		M	F
2200	222		M	F
2700	272		M	G
3300	332		M	G/F
3900	392		C	G
4700	472		C	G
5600	562	E	C	G
6800	682	C/E	E	G
8200	822	E	E	G
10000	103	E	F	G
12000	123		F	G
15000	153		G	G
18000	183		G	G
22000	223		G	G
27000	273		G	
33000	333		G	
39000	393			
47000	473			
56000	563			
68000	683			
82000	823			
100000	104			

* Surface coating only.



7. CAPACITANCE RANGE

7-2 X7R

DIELECTRIC		X7R									
SIZE		0402	0603			0805					
RATED VOLTAGE (V)		16V	10V 16V 25V	50V	100V	10V	16V	25V	50V	100V	250V 630V
Capacitance	100pF (101)										
	120pF (121)										
	150pF (151)										
	180pF (181)										
	220pF (221)					X	X	X	X	X	X
	270pF (271)					X	X	X	X	X	X
	330pF (331)					X	X	X	X	X	X
	390pF (391)					X	X	X	X	X	X
	470pF (471)					X	X	X	X	X	X
	560pF (561)					X	X	X	X	X	X
	680pF (681)					X	X	X	X	X	X
	820pF (821)					X	X	X	X	X	X
	1,000pF (102)		S	S	S	X/C	X/C	X/C	X/C	X	X
	1,200pF (122)		S	S	S	X/C	X/C	X/C	X/C	X	X
	1,500pF (152)		S	S	S	X/C	X/C	X/C	X/C	X	X
	1,800pF (182)		S	S	S	X/C	X/C	X/C	X/C	X	X
	2,200pF (222)		S	S	S	X/C	X/C	X/C	X/C	X	X
	2,700pF (272)		S	S	S	X/C	X/C	X/C	X/C	X	X
	3,300pF (332)		S	S	S	X/C	X/C	X/C	X/C	X	X
	3,900pF (392)		S	S	S	X/C	X/C	X/C	X/C	X	X
	4,700pF (472)		S	S	S	X/C	X/C	X/C	X/C	X	X
	5,600pF (562)		S	S	S	X/C	X/C	X/C	X/C	X	M
	6,800pF (682)		S	S	S	X/C	X/C	X/C	X/C	X	C
	8,200pF (822)		S	S	S	X/C	X/C	X/C	X/C	X	C
	0.010μF (103)		S	S	S	X/C	X/C	X/C	X/C	X	C
	0.012μF (123)		S	S	S	X/C	X/C	X/C	X/C	X	C
	0.015μF (153)		S	S	S	X/C	X/C	X/C	X/C	X	C
	0.018μF (183)		S	S	S	X/C	X/C	X/C	X/C	X	C
	0.022μF (223)		S	S	S	X/C	X/C	X/C	X/C	X	C
	0.027μF (273)		S	S	S	X/C	X/C	M/C	M/C	M	
	0.033μF (333)		S	B	S	X/C	X/C	M/C	M/C	M	
	0.039μF (393)		S	B	S	X/C	X/C	M/C	M/C	M	
	0.047μF (473)		S	B	S	X/C	X/C	M/C	M/C	M	
	0.056μF (563)	K	S	B	S	X/C	X/C	M/C	M/C	M	
	0.068μF (683)	K	S	B	S	X/C	X/C	M/C	M/C	M	
	0.082μF (823)	K	S	B	S	X/C	X/C	M/C	M/C	M	
	0.10μF (104)	K	S	B	B	X/C	X/C	C	C	C	
	0.12μF (124)		B			X	X	C	C	C	
	0.15μF (154)		B			X	X	C	C	C	
	0.18μF (184)		B			X	X	C	C	C	
	0.22μF (224)		B			X	X	C	C	C	
	0.27μF (274)					C	C	C	C		
	0.33μF (334)					C	C	C	C		
	0.39μF (394)					C	C	C	C		
	0.47μF (474)					C	C	C	C		
	0.56μF (564)					C	C	C/I	C/I		
	0.68μF (684)					C	C	C/I	C/I		
	0.82μF (824)					C	C	C/I	C/I		
	1.0μF (105)					C	C	C/I	C/I		
	1.2μF (125)										
	1.5μF (155)										
	1.8μF (185)										
	2.2μF (225)										
	2.7μF (275)										
	3.3μF (335)										
	3.9μF (395)										
	4.7μF (475)										

* Surface coating only.



7. CAPACITANCE RANGE

7-2 X7R

DIELECTRIC		X7R									
SIZE		1206									
RATED VOLTAGE (V)		10V	16V	25V	50V	100V	250V	500V	630V	1000V	2000V
Capacitance	100pF (101)										
	120pF (121)										
	150pF (151)										
	180pF (181)										
	220pF (221)	X	X	X	X	X	X	X	X	X	
	270pF (271)	X	X	X	X	X	X	X	X	X	
	330pF (331)	X	X	X	X	X	X	X	X	X	
	390pF (391)	X	X	X	X	X	X	X	X	X	
	470pF (471)	X	X	X	X	X	X	X	X	X	C*
	560pF (561)	X	X	X	X	X	X	X	X	X	C*
	680pF (681)	X	X	X	X	X	X	X	X	M	C*
	820pF (821)	X	X	X	X	X	X	X	X	M	C*
	1,000pF (102)	X	X	X	X	X	X	X	X	C	C*
	1,200pF (122)	X	X	X	X	X	X	X	X	E	C*
	1,500pF (152)	X	X	X	X	X	X	X	X	E	C*
	1,800pF (182)	X	X	X	X	X	X	X	X	E	C*
	2,200pF (222)	X	X	X	X	X	X	X	X	E	C*
	2,700pF (272)	X	X	X	X	X	X	X	X	E	C*
	3,300pF (332)	X	X	X	X	X	X	X	X	E	E*
	3,900pF (392)	X	X	X	X	X	X	X	X	E	E*
	4,700pF (472)	X	X	X	X	X	X	X	X	E	E*
	5,600pF (562)	X	X	X	X	X	X	X	X	E	
	6,800pF (682)	X	X	X	X	X	X	M	M	E	
	8,200pF (822)	X	X	X	X	X	X	M	M	E	
	0.010μF (103)	X	X	X	X	X	M	M	M	E	
	0.012μF (123)	X	X	X	X	X	C	C	C		
	0.015μF (153)	X	X	X	X	X	C	E	E		
	0.018μF (183)	X	X	X	X	X	C	E	E		
	0.022μF (223)	X	X	X	X	X	E	E	E		
	0.027μF (273)	X	X	X	X	X	E				
	0.033μF (333)	X	X	X	X	X	E				
	0.039μF (393)	X	X	X	X	X	E				
	0.047μF (473)	X	X	X	X	X	E				
	0.056μF (563)	X	X	X	X	M					
	0.068μF (683)	X	X	X	X	C					
	0.082μF (823)	X	X	X	X	C					
	0.10μF (104)	X	X	X	X	E					
	0.12μF (124)	X	X	X	X	P					
	0.15μF (154)	X	X	X	X	P					
	0.18μF (184)	X	X	M	M	P					
	0.22μF (224)	X	X	M	M	E/P					
	0.27μF (274)	X	X	C	C	P					
	0.33μF (334)	X	X	C	C	P					
	0.39μF (394)	C	C	E	E	P					
	0.47μF (474)	C	C	E	E	P					
	0.56μF (564)	C	C	P	P	P					
	0.68μF (684)	C	C	P	P	P					
	0.82μF (824)	E	E	P	P	P					
	1.0μF (105)	P	P	P	P	P					
	1.2μF (125)	P	P	P	P	P					
	1.5μF (155)	P	P	P	P	P					
	1.8μF (185)	P	P	P	P	P					
	2.2μF (225)	P	P	P	P	P					
	2.7μF (275)	P	P	P	P						
	3.3μF (335)	P	P	P	P						
	3.9μF (395)	P	P	P	P						
	4.7μF (475)	P	P	P	P						
	5.6μF (565)	P	P	P							
	6.8μF (685)	P	P	P							
	8.2μF (825)	P	P	P							
	10μF (106)	P	P	P							

* Surface coating only.



7. CAPACITANCE RANGE

7-2 X7R

DIELECTRIC SIZE		X7R 1210							
RATED VOLTAGE (V)		10V	16V	25V	50V	100V	250V	630V	1000V
Capacitance	100pF (101)								
	120pF (121)								
	150pF (151)								
	180pF (181)								
	220pF (221)	M	M	M	M	M	M	M	M
	270pF (271)	M	M	M	M	M	M	M	M
	330pF (331)	M	M	M	M	M	M	M	M
	390pF (391)	M	M	M	M	M	M	M	M
	470pF (471)	M	M	M	M	M	M	M	M
	560pF (561)	M	M	M	M	M	M	M	M
	680pF (681)	M	M	M	M	M	M	M	M
	820pF (821)	M	M	M	M	M	M	M	M
	1,000pF (102)	M	M	M	M	M	M	M	C
	1,200pF (122)	M	M	M	M	M	M	M	E
	1,500pF (152)	M	M	M	M	M	M	M	E
	1,800pF (182)	M	M	M	M	M	M	M	E
	2,200pF (222)	M	M	M	M	M	M	M	F
	2,700pF (272)	M	M	M	M	M	M	M	F
	3,300pF (332)	M	M	M	M	M	M	M	F
	3,900pF (392)	M	M	M	M	M	M	M	G
	4,700pF (472)	M	M	M	M	M	M	M	G
	5,600pF (562)	M	M	M	M	M	M	C	G
	6,800pF (682)	M	M	M	M	M	M	C	G
	8,200pF (822)	M	M	M	M	M	M	G	G
	0.010μF (103)	M	M	M	M	M	M	G	G
	0.012μF (123)	M	M	M	M	M	M	G	G
	0.015μF (153)	M	M	M	M	M	M	G	G
	0.018μF (183)	M	M	M	M	M	C	E	G
	0.022μF (223)	M	M	M	M	M	C	E	G
	0.027μF (273)	M	M	M	M	M	C	E	
	0.033μF (333)	M	M	M	M	M	E	E	
	0.039μF (393)	M	M	M	M	M	E	F	
	0.047μF (473)	M	M	M	M	M	E	G	
	0.056μF (563)	M	M	M	M	M	E		
	0.068μF (683)	M	M	M	M	M	F		
	0.082μF (823)	M	M	M	M	M	F		
	0.10μF (104)	M	M	M	M	M	F		
	0.12μF (124)	M	M	C	C	E			
	0.15μF (154)	M	M	C	C	E			
	0.18μF (184)	M	M	E	E	E			
	0.22μF (224)	M	M	E	E	E			
	0.27μF (274)	M	M	M	M	F			
	0.33μF (334)	M	M	M	M	F			
	0.39μF (394)	M	M	F	F	G			
	0.47μF (474)	M	M	F	F	G			
	0.56μF (564)	M	M	F	F	G			
	0.68μF (684)	M	M	F	F	G			
	0.82μF (824)	C	C	F	F	G			
	1.0μF (105)	C	C	F	F	G			
	1.2μF (125)	F	F	F	F	G			
	1.5μF (155)	F	F	F	F	G			
	1.8μF (185)	G	G	G	G	G			
	2.2μF (225)	G	G	G	G	G			
	2.7μF (275)	G	G	G	G	G			
	3.3μF (335)	G	G	G	G	G			
	3.9μF (395)	G	G	G	G	G			
	4.7μF (475)	G	G	G	G	G			
	5.6μF (565)	G	G	G	G				
	6.8μF (685)	G	G	G	G				
	8.2μF (825)	G	G	G	G				
	10μF (106)	G	G	G	G				
	12.0μF (126)	G	G	G					
	15.0μF (156)	G	G	G					
	18.0μF (186)	G	G	G					
	22.0μF (226)	G	G	G					

* Surface coating only.



7. CAPACITANCE RANGE

7-2 X7R

DIELECTRIC		X7R										
SIZE		1808	1812									
RATED VOLTAGE (V)		2000V	10V	16V	25V	50V	100V	250V	630V	1000V	2000V	3000V
Capacitance	100pF (101)											
	120pF (121)											
	150pF (151)	C										
	180pF (181)	C										
	220pF (221)	C										
	270pF (271)	C	C	C	C	C	C	C	C	C	C	F*
	330pF (331)	C	C	C	C	C	C	C	C	C	C	F*
	390pF (391)	C	C	C	C	C	C	C	C	C	C	F*
	470pF (471)	C	C	C	C	C	C	C	C	C	C	F*
	560pF (561)	E	C	C	C	C	C	C	C	C	C	F*
	680pF (681)	E	C	C	C	C	C	C	C	C	C	F*
	820pF (821)	E	C	C	C	C	C	C	C	C	C	F*
	1,000pF (102)	F	C	C	C	C	C	C	C	C	E	F*
	1,200pF (122)		C	C	C	C	C	C	C	C	F	G*
	1,500pF (152)		C	C	C	C	C	C	C	C	F	G*
	1,800pF (182)		C	C	C	C	C	C	C	C	G	G*
	2,200pF (222)		C	C	C	C	C	C	C	C	G*	
	2,700pF (272)		C	C	C	C	C	C	C	C	G*	
	3,300pF (332)		C	C	C	C	C	C	C	E	G*	
	3,900pF (392)		C	C	C	C	C	C	C	F	G*	
	4,700pF (472)		C	C	C	C	C	C	C	F	G*	
	5,600pF (562)		C	C	C	C	C	C	C	G	G*	
	6,800pF (682)		C	C	C	C	C	C	C	G	G*	
	8,200pF (822)		C	C	C	C	C	C	C	G	G*	
	0.010μF (103)		C	C	C	C	C	C	C	G	G*	
	0.012μF (123)		C	C	C	C	C	C	C			
	0.015μF (153)		C	C	C	C	C	C	C			
	0.018μF (183)		C	C	C	C	C	C	E			
	0.022μF (223)		C	C	C	C	C	C	E			
	0.027μF (273)		C	C	C	C	C	C	F			
	0.033μF (333)		C	C	C	C	C	C	F			
	0.039μF (393)		C	C	C	C	C	C	G			
	0.047μF (473)		C	C	C	C	C	C	G			
	0.056μF (563)		C	C	C	C	C	E	G			
	0.068μF (683)		C	C	C	C	C	E	G			
	0.082μF (823)		C	C	C	C	C	E	G			
	0.10μF (104)		C	C	C	C	C	E	G			
	0.12μF (124)		C	C	C	C	C	F				
	0.15μF (154)		C	C	C	C	C	F				
	0.18μF (184)		C	C	C	C	C	G				
	0.22μF (224)		C	C	C	C	E	G				
	0.27μF (274)		C	C	C	C	E	G				
	0.33μF (334)		C	C	C	C	F	G				
	0.39μF (394)		C	C	C	C	F	G				
	0.47μF (474)		C	C	C	C	G	G				
	0.56μF (564)		C	C	C	C	G					
	0.68μF (684)		C	C	C	C	G					
	0.82μF (824)		C	C	C	C	G					
	1.0μF (105)		C	C	C	C	G					
	1.2μF (125)		C	C	C	C						
	1.5μF (155)		C	C	C	C						
	1.8μF (185)		E	E	E	E						
	2.2μF (225)		E	E	E	E						
	2.7μF (275)		F	F	F	F						
	3.3μF (335)		F	F	F	F						
	3.9μF (395)		F	F	F	F						
	4.7μF (475)		G	G	G	G						
	5.6μF (565)		G	G								
	6.8μF (685)		G	G								
	8.2μF (825)		G	G								
	10.0μF (106)		G	G								

* Surface coating only



7. CAPACITANCE RANGE

7-2 X7R

DIELECTRIC		X7R									
SIZE		2220									
RATED VOLTAGE (V)		10V	16V	25V	50V	100V	250V	630V	1000V	2000V	3000V
Capacitance	100pF (101)										
	120pF (121)										
	150pF (151)										
	180pF (181)										
	220pF (221)										
	270pF (271)										F*
	330pF (331)										F*
	390pF (391)										F*
	470pF (471)										F*
	560pF (561)										F*
	680pF (681)										F*
	820pF (821)										F*
	1,000pF (102)	F	F	F	F	F	F	F	F	F	F*
	1,200pF (122)	F	F	F	F	F	F	F	F	F	G*
	1,500pF (152)	F	F	F	F	F	F	F	F	F	G*
	1,800pF (182)	F	F	F	F	F	F	F	F	F	G*
	2,200pF (222)	F	F	F	F	F	F	F	F	F*	G*
	2,700pF (272)	F	F	F	F	F	F	F	F	F*	G*
	3,300pF (332)	F	F	F	F	F	F	F	F	F*	G*
	3,900pF (392)	F	F	F	F	F	F	F	F	F*	G*
	4,700pF (472)	F	F	F	F	F	F	F	F	F*	G*
	5,600pF (562)	F	F	F	F	F	F	F	F	F*	G*
	6,800pF (682)	F	F	F	F	F	F	F	F	G*	G*
	8,200pF (822)	F	F	F	F	F	F	F	G	G*	G*
	0.010μF (103)	F	F	F	F	F	F	F	G	G*	G*
	0.012μF (123)	F	F	F	F	F	F	F	G	H*	
	0.015μF (153)	F	F	F	F	F	F	F	G	H*	
	0.018μF (183)	F	F	F	F	F	F	F	H	H*	
	0.022μF (223)	F	F	F	F	F	F	F	H		
	0.027μF (273)	F	F	F	F	F	F	F	H		
	0.033μF (333)	F	F	F	F	F	F	F	H		
	0.039μF (393)	F	F	F	F	F	F	F	H		
	0.047μF (473)	F	F	F	F	F	F	F	H		
	0.056μF (563)	F	F	F	F	F	F	F	H		
	0.068μF (683)	F	F	F	F	F	F	F	F		
	0.082μF (823)	F	F	F	F	F	F	G	F		
	0.10μF (104)	F	F	F	F	F	F	G	F		
	0.12μF (124)	F	F	F	F	F	F	H			
	0.15μF (154)	F	F	F	F	F	F	H			
	0.18μF (184)	F	F	F	F	F	F	H			
	0.22μF (224)	F	F	F	F	F	F	H			
	0.27μF (274)	F	F	F	F	F	F				
	0.33μF (334)	F	F	F	F	F	F				
	0.39μF (394)	F	F	F	F	F	F				
	0.47μF (474)	F	F	F	F	F	F				
	0.56μF (564)	F	F	F	F	F					
	0.68μF (684)	F	F	F	F	F					
	0.82μF (824)	F	F	F	F	F					
	1.0μF (105)	F	F	F	F	F					
	1.2μF (125)	F	F	F	F	G					
	1.5μF (155)	F	F	F	F	G					
	1.8μF (185)	F	F	F	F	G					
	2.2μF (225)	F	F	F	F	G					
	2.7μF (275)	F	F	F	F	G					
	3.3μF (335)	F	F	F	F	G					
	3.9μF (395)	F	F	F	F	G					
	4.7μF (475)	F	F	F	F	G					
	5.6μF (565)	F	F	F	F	G					
	6.8μF (685)	F	F	F	F	G					
	8.2μF (825)	G	G	G	G	G					
	10.0μF (106)	G	G	G	G	G					
	12.0μF (126)	H	H			G					
	15.0μF (156)	H	H			G					
	18.0μF (186)	H	H								
	22.0μF (226)	H	H								

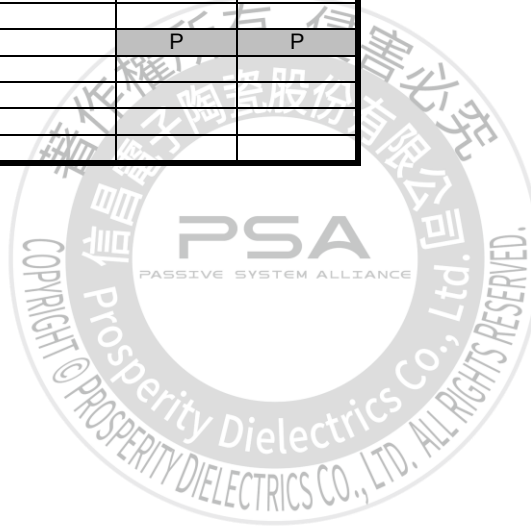
* Surface coating only.



7. CAPACITANCE RANGE

7-3 X7S

DIELECTRIC		X7S			
SIZE		0805		1206	
RATED VOLTAGE (V)		10	16	10	16
Capacitance	0.10μF (104)				
	0.12μF (124)				
	0.15μF (154)				
	0.18μF (184)				
	0.22μF (224)				
	0.27μF (274)				
	0.33μF (334)				
	0.39μF (394)				
	0.47μF (474)				
	0.56μF (564)				
	0.68μF (684)				
	0.82μF (824)	C	C		
	1.0μF (105)	C	C		
	1.2μF (125)				
	1.5μF (155)				
	1.8μF (185)				
	2.2μF (225)			P	P
	2.7μF (275)				
	3.3μF (335)				
	3.9μF (395)				
	4.7μF (475)				



8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements																																													
1	Pre-and Post-Stress Electrical Test	---	---																																													
2	High Temperature Exposure (Storage) MIL-STD-202 Method 108	* Test temp. : 150±3°C. * Unpowered. * Test time : 1000 +24/-0 hrs. * Measurement to be made after keeping at room temp. for 24±2 hrs.	* No remarkable damage. * Cap. change : C0G within ±2.5% or ±0.25pF, whichever is larger. X7R/X7S within ±10.0%. * D.F. value :C0G : To meet the initial requirement. X7R/X7S : D.F.≤200% of initial requirement. * X7R/X7S : <table><tr><th>Rated vol.</th><th>D.F.≤</th><th>Exception of D.F.≤</th></tr><tr><td rowspan="3">≥100V</td><td rowspan="3">≤3.5%</td><td>≤6% 1206≥0.47μF</td></tr><tr><td>≤7.5% 0603≥0.068μF;0805>0.1μF;1206≥1μF;1210≥2.2μF</td></tr><tr><td>≤20% 0805>0.22μF;1210≥3.3μF</td></tr><tr><td rowspan="3">50V</td><td rowspan="3">≤3.5%</td><td>≤6% 0201(50V);0603≥0.047μF;0805≥0.18μF;1206≥0.47μF,</td></tr><tr><td>≤10% 0201≥0.01uF;1210≥3.3μF</td></tr><tr><td>≤20% 0402≥0.012μF;0603>0.1μF; 0805/X7R>0.47μF; 1206≥2.2μF;1210≥10μF</td></tr><tr><td>35V</td><td>≤5%</td><td>≤20% 0603≥1μF;0805≥2.2μF;1206≥2.2μF;1210≥10μF</td></tr><tr><td rowspan="4">25V</td><td rowspan="4">≤5%</td><td>≤10% 0201≥0.01μF;0805≥1μF;1210≥10μF</td></tr><tr><td>≤14% 0603≥0.33μF</td></tr><tr><td>≤15% 0201≥0.1μF; 0402≥0.056μF;0603≥0.47μF; 0805≥2.2μF;1206≥4.7μF;1210≥22μF</td></tr><tr><td>≤20% 0402≥0.47μF</td></tr><tr><td rowspan="2">16V</td><td rowspan="2">≤5%</td><td>≤10% 0603≥0.15μF;0805≥0.68μF;1206≥2.2μF;1210≥4.7μF</td></tr><tr><td>≤15% 0201≥0.022μF;0402≥0.033μF; 0603>0.47μF;0805≥2.2μF;1206≥4.7μF;1210≥22μF</td></tr><tr><td rowspan="2">10V</td><td rowspan="2">≤7.5%</td><td>≤15% 0201≥0.012μF;0402≥0.22μF; 0603≥0.33μF;0805≥2.2μF;1206≥2.2μF;1210≥22μF</td></tr><tr><td>≤20% 0201≥0.1μF;0402≥1μF</td></tr><tr><td>6.3V</td><td>≤15%</td><td>≤30% 0201≥0.1μF;0402≥1μF;0603≥10μF; 0805≥4.7μF;1206≥47μF;1210≥100μF</td></tr><tr><td>4V</td><td>≤20%</td><td>---</td></tr></table> * I.R. : To meet the initial requirement. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="7">≥1GΩ or RxC≥10Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated vol.	D.F.≤	Exception of D.F.≤	≥100V	≤3.5%	≤6% 1206≥0.47μF	≤7.5% 0603≥0.068μF;0805>0.1μF;1206≥1μF;1210≥2.2μF	≤20% 0805>0.22μF;1210≥3.3μF	50V	≤3.5%	≤6% 0201(50V);0603≥0.047μF;0805≥0.18μF;1206≥0.47μF,	≤10% 0201≥0.01uF;1210≥3.3μF	≤20% 0402≥0.012μF;0603>0.1μF; 0805/X7R>0.47μF; 1206≥2.2μF;1210≥10μF	35V	≤5%	≤20% 0603≥1μF;0805≥2.2μF;1206≥2.2μF;1210≥10μF	25V	≤5%	≤10% 0201≥0.01μF;0805≥1μF;1210≥10μF	≤14% 0603≥0.33μF	≤15% 0201≥0.1μF; 0402≥0.056μF;0603≥0.47μF; 0805≥2.2μF;1206≥4.7μF;1210≥22μF	≤20% 0402≥0.47μF	16V	≤5%	≤10% 0603≥0.15μF;0805≥0.68μF;1206≥2.2μF;1210≥4.7μF	≤15% 0201≥0.022μF;0402≥0.033μF; 0603>0.47μF;0805≥2.2μF;1206≥4.7μF;1210≥22μF	10V	≤7.5%	≤15% 0201≥0.012μF;0402≥0.22μF; 0603≥0.33μF;0805≥2.2μF;1206≥2.2μF;1210≥22μF	≤20% 0201≥0.1μF;0402≥1μF	6.3V	≤15%	≤30% 0201≥0.1μF;0402≥1μF;0603≥10μF; 0805≥4.7μF;1206≥47μF;1210≥100μF	4V	≤20%	---	Rated voltage	I.R.	100V : All X7R/X7S	≥1GΩ or RxC≥10Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF
Rated vol.	D.F.≤	Exception of D.F.≤																																														
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Rated voltage	I.R.																																															
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3		Destructive Physical Analysis EIA-469	Per EIA-469.	* No defects or abnormalities.																																												
4	Temperature Cycling JESD22 Method JA-104	* Conduct 1000 cycles according to the temperatures and time. <table><tr><th>Step</th><th>Temp.(°C)</th><th>Time(min.)</th></tr><tr><td>1</td><td>-55°C +0/-3</td><td>15±1</td></tr><tr><td>2</td><td>+125°C +3/-0</td><td>15±1</td></tr></table> * Before initial measurement : Perform 150 +0/-10°C for 1 hr and then set for 24±2 hrs at room temp. * Measurement to be made after keeping at room temp. for 24±2 hrs.	Step	Temp.(°C)	Time(min.)	1	-55°C +0/-3	15±1	2	+125°C +3/-0	15±1	* No remarkable damage. * Cap. change : C0G : Within ±2.5% or ±0.25pF, whichever is larger. X7R/X7S within ±10.0%. * D.F. value :.C0G : To meet the initial requirement. X7R/X7S : D.F.≤200% of initial requirement. * I.R. : To meet the initial requirement. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">≥1GΩ or RxC≥10Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	≥1GΩ or RxC≥10Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF																											
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8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements														
5	Moisture Resistance MIL-STD-202 Method 106	* Test temp. : 25~65°C. * Humidity : 80~100% RH. * Test time : 10 cycles, t=24hrs/cycle. * Measurement to be made after keeping at room temp. for 24±2 hrs.	* No remarkable damage. * Cap. change : C0G : Within ±3.0% or ±0.3pF, whichever is larger. X7R/X7S within ±12.5%. * D.F. value :. C0G : Q≥350 for Cap.>30pF. Q≥275+2.5C for 10pF≤Cap.≤30pF. Q≥200+10C for Cap.<10pF. X7R/X7S : D.F.≤200% of initial requirement. * I.R. : ≥1GΩ or RxC≥10Ω-F, whichever is smaller. Class II (X7R/X7S)														
			Rated voltage	I.R.	100V : All X7R/X7S	≥1GΩ or RxC≥10Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF						
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100V : All X7R/X7S	≥1GΩ or RxC≥10Ω-F, whichever is smaller																
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6		Biased Humidity MIL-STD-202 Method 103	* Test temp. : 85±3°C. * Humidity : 85±5%RH. * Test time : 1000 +24/-0 hrs. * To apply voltage : Rated voltage (max. 100Vdc) and 1.3~1.5Vdc (add 100k ohm resistor). * Before initial measurement (Class II only) : To apply test voltage for 1hr at test temp. and then set for 24±2 hrs at room temp. * Measurement to be made after keeping at room temp. for 24±2 hrs.	* No remarkable damage. * Cap. change : C0G : Within ±3.0% or ±0.3pF, whichever is larger. X7R/X7S within ±12.5%. * D.F. value : C0G : Q≥200 for Cap.≥30pF. Q≥100+10/3C for Cap.<30pF. X7R/X7S : D.F.≤200% of initial requirement. * I.R. : ≥500MΩ or RxC≥25Ω-F, whichever is smaller. Class II (X7R/X7S) for rated voltage test													
	Rated voltage			I.R.	100V : All X7R/X7S	≥500MΩ or RxC≥5Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF	Rated voltage	I.R.	100V : All X7R/X7S	≥1GΩ or RxC≥10Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF
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8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements									
7	Operational Life MIL-STD-202 Method 108	* Test temp. : 125±3°C. * To apply voltage : Full rated voltage. * Test time : 1000 +24/-0 hrs. * Before initial measurement : Apply rated voltage for 1 hr at 125°C. Remove and let set for 24±2 hrs at room temp. * Measurement to be made after keeping at room temp. for 24±2 hrs.	* No remarkable damage. * Cap. change : C0G : Within ±3.0% or ±0.3pF, whichever is larger. X7R/X7S within ±12.5%. * D.F. value : C0G: Q≥350 for Cap.>30pF, Q≥275+2.5C for 10pF≤Cap.≤30pF, Q≥200+10C for Cap.<10pF. X7R/X7S : D.F.≤200% of initial requirement. * I.R. : ≥1GΩ or RxC≥50Ω-F, whichever is smaller. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">≥1GΩ or RxC≥100Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	≥1GΩ or RxC≥100Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥4.7μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF
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8	External Visual MIL-STD-883 Method 2009	* Visual inspection.	* No remarkable defect.									
9	Physical Dimension JESD22 Method JB-100	* Using by calipers.	* Within the specified dimensions.									
10	Resistance to Solvents MIL-STD-202 Method 215	* Temperature : 25±5°C. * Time : 3 +0.5/-0 min. * Solvent : Iso-propyl alcohol.	* No remarkable damage. * Cap. : Within the specified tolerance. * Q/D.F. value : To meet the initial requirement. * I.R. : To meet the initial requirement. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">≥10GΩ or RxC≥100Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	≥10GΩ or RxC≥100Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF
Rated voltage	I.R.											
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8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements						
11	Mechanical Shock MIL-STD-202 Method 213	* Peak value : 1500g's. * Wave : 1/2 sine. * Velocity : 15.4 ft/sec. * Three shocks in each direction should be applied along 3 mutually perpendicular axes of the test specimen (18 shocks).	* No remarkable damage. * Cap. : Within the specified tolerance. * D.F. value :. To meet the initial requirement. * I.R. : $\geq 10G\Omega$ or $RxC\geq 500\Omega\cdot F$, whichever is smaller. Class II (X7R/X7S)						
			<table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">$\geq 10G\Omega$ or $RxC\geq 100\Omega\cdot F$, whichever is smaller</td></tr><tr><td>50V : 0402$\geq 0.01\mu F$, 0603$\geq 1\mu F$, 0805$\geq 1\mu F$, 1206$\geq 1.0\mu F$, 1210$\geq 4.7\mu F$</td></tr><tr><td>35V : 0805$\geq 2.2\mu F$, 1206$\geq 2.2\mu F$, 1210$\geq 10\mu F$</td></tr><tr><td>25V : 0402$\geq 1\mu F$, 0603$\geq 2.2\mu F$, 0805$\geq 2.2\mu F$, 1206$\geq 1\mu F$, 1210$\geq 10\mu F$</td></tr><tr><td>16V : 0402$\geq 0.22\mu F$, 0603$\geq 1\mu F$, 0805$\geq 2.2\mu F$, 1206$\geq 10\mu F$, 1210$\geq 47\mu F$</td></tr><tr><td>10V : 0402$\geq 0.47\mu F$, 0603$\geq 0.47\mu F$, 0805$\geq 2.2\mu F$, 1206$\geq 4.7\mu F$, 1210$\geq 47\mu F$</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	$\geq 10G\Omega$ or $RxC\geq 100\Omega\cdot F$, whichever is smaller	50V : 0402 $\geq 0.01\mu F$, 0603 $\geq 1\mu F$, 0805 $\geq 1\mu F$, 1206 $\geq 1.0\mu F$, 1210 $\geq 4.7\mu F$	35V : 0805 $\geq 2.2\mu F$, 1206 $\geq 2.2\mu F$, 1210 $\geq 10\mu F$
Rated voltage	I.R.								
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10V : 0402 $\geq 0.47\mu F$, 0603 $\geq 0.47\mu F$, 0805 $\geq 2.2\mu F$, 1206 $\geq 4.7\mu F$, 1210 $\geq 47\mu F$									
12	Vibration MIL-STD-202 Method 204	* Vibration frequency : 10~2000 Hz/min. (5g's for 20 min.). * Total amplitude : 1.5mm. * 12 cycles each of 3 orientations (36 times).	* No remarkable damage. * Cap. : Within the specified tolerance. * D.F. value :. To meet the initial requirement. * I.R. : $\geq 10G\Omega$ or $RxC\geq 500\Omega\cdot F$, whichever is smaller. Class II (X7R/X7S)						
			<table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">$\geq 10G\Omega$ or $RxC\geq 100\Omega\cdot F$, whichever is smaller</td></tr><tr><td>50V : 0402$\geq 0.01\mu F$, 0603$\geq 1\mu F$, 0805$\geq 1\mu F$, 1206$\geq 1.0\mu F$, 1210$\geq 4.7\mu F$</td></tr><tr><td>35V : 0805$\geq 2.2\mu F$, 1206$\geq 2.2\mu F$, 1210$\geq 10\mu F$</td></tr><tr><td>25V : 0402$\geq 1\mu F$, 0603$\geq 2.2\mu F$, 0805$\geq 2.2\mu F$, 1206$\geq 1\mu F$, 1210$\geq 10\mu F$</td></tr><tr><td>16V : 0402$\geq 0.22\mu F$, 0603$\geq 1\mu F$, 0805$\geq 2.2\mu F$, 1206$\geq 10\mu F$, 1210$\geq 47\mu F$</td></tr><tr><td>10V : 0402$\geq 0.47\mu F$, 0603$\geq 0.47\mu F$, 0805$\geq 2.2\mu F$, 1206$\geq 4.7\mu F$, 1210$\geq 47\mu F$</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	$\geq 10G\Omega$ or $RxC\geq 100\Omega\cdot F$, whichever is smaller	50V : 0402 $\geq 0.01\mu F$, 0603 $\geq 1\mu F$, 0805 $\geq 1\mu F$, 1206 $\geq 1.0\mu F$, 1210 $\geq 4.7\mu F$	35V : 0805 $\geq 2.2\mu F$, 1206 $\geq 2.2\mu F$, 1210 $\geq 10\mu F$
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8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements									
13	Resistance to Soldering Heat MIL-STD-202 Method 210	* Solder temperature : 260±5°C. * Dipping time : 10±1 sec. * Before initial measurement : Perform 150 +0/-10°C for 1 hr and then set for 24±2 hrs at room temp. * Measurement to be made after keeping at room temp. for 24±2 hrs.	* No remarkable damage. * Cap. change : X7R/X7S within ±7.5%. * D.F. value : C0G : Within ±2.5% or ±0.25pF, whichever is larger. X7R/X7S : D.F.≤100% of initial requirement. * I.R. : ≥10GΩ or RxC≥500Ω-F, whichever is smaller. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">≥10GΩ or RxC≥100Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥1μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	≥10GΩ or RxC≥100Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥1μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF
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50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF												
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25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥1μF, 1210≥10μF												
16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF												
10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF												
14	ESD AEC-Q200-002	* Per AEC-Q200-002.	* No remarkable damage. * Cap. : Within the specified tolerance. * D.F. value : To meet the initial requirement. X7R/X7S : D.F.≤100% of initial requirement. * I.R. : To meet the initial requirement.. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">≥10GΩ or RxC≥100Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥1μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	≥10GΩ or RxC≥100Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF	35V : 0805≥2.2μF, 1206≥1μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥10μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF
Rated voltage	I.R.											
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16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF												
10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF												
15	Solderability J-STD-002 JESD22-B102E	* Condition A Un-mounted chips 4hrs / 155°C*dry then completely immersed for 5±0.5 sec in solder bath at 245±5°C. * Condition B Un-mounted chips steam 8 hrs then completely immersed for 10±1sec in solder bath at 220 +5/-0°C. * Condition C Un-mounted chips steam 8 hrs then completely immersed for 10±1 sec. in solder bath at 260 +0/-5°C.	* All terminations shall exhibit a continuous solder coating free from defects from a minimum of 95% of the critical surface area of any individual termination.									



8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements																																																	
16	Electrical Characterization		* Capacitance within the specified tolerance. * D.F. value : To meet the initial requirement. X7R/X7S : <table><tr><th>Rated</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr><tr><td rowspan="3">≥100V</td><td rowspan="3">≤2.5%</td><td>≤3.5%</td><td>1206≥0.47μF, 1812≥4.7μF, 1825≥4.7μF, 2220≥4.7μF, 2225≥4.7μF</td></tr><tr><td>≤5%</td><td>0805>0.1μF, 0603≥0.068μF, 1206>1μF, 1210≥2.2μF</td></tr><tr><td>≤10%</td><td>0805>0.22μF, 1210≥3.3μF</td></tr><tr><td rowspan="3">50V</td><td rowspan="3">≤2.5%</td><td>≤3.5%</td><td>0201(50V), 0603≥0.047μF, 0805≥0.18μF, 1206≥0.47μF, 1210≥3.3μF, 1812≥4.7μF, 1825≥4.7μF, 2220≥4.7μF, 2225≥4.7μF</td></tr><tr><td>≤5%</td><td>0201≥0.01μF, 1210≥4.7μF</td></tr><tr><td>≤10%</td><td>0402≥0.1μF, 0603>0.1μF, 0805≥1μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>35V</td><td>≤3.5%</td><td>≤10%</td><td>0603≥1μF, 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td rowspan="3">25V</td><td rowspan="3">≤3.5%</td><td>≤5%</td><td>0201≥0.01μF, 0805≥1μF, 1210≥10μF</td></tr><tr><td>≤7%</td><td>0603≥0.33μF, 1206≥4.7μF</td></tr><tr><td>≤10%</td><td>0201≥0.1μF, 0402≥0.10μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥6.8μF, 1210≥22μF</td></tr><tr><td rowspan="3">16V</td><td rowspan="3">≤3.5%</td><td>≤12.5%</td><td>0402≥0.47μF</td></tr><tr><td>≤5%</td><td>0201≥0.01μF, 0402≥0.033μF, 0603≥0.15μF, 0805≥0.68μF, 1206≥2.2μF, 1210≥4.7μF</td></tr><tr><td>≤10%</td><td>0201≥0.1μF(0201/X7R/X7S≥0.022μF), 0402≥0.22μF, 0603≥0.68μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥22μF</td></tr><tr><td>10V</td><td>≤5%</td><td>≤10%</td><td>0201≥0.012μF, 0402≥0.33μF(0402/X7R/X7S≥0.22μF), 0603≥0.33μF, 0805≥2.2μF, 1206≥2.2μF, 1210≥22μF</td></tr><tr><td></td><td></td><td></td><td>≤15%</td><td>0201≥0.1μF, 0402≥1μF</td></tr></table>	Rated	D.F.≤	Exception of D.F.≤		≥100V	≤2.5%	≤3.5%	1206≥0.47μF, 1812≥4.7μF, 1825≥4.7μF, 2220≥4.7μF, 2225≥4.7μF	≤5%	0805>0.1μF, 0603≥0.068μF, 1206>1μF, 1210≥2.2μF	≤10%	0805>0.22μF, 1210≥3.3μF	50V	≤2.5%	≤3.5%	0201(50V), 0603≥0.047μF, 0805≥0.18μF, 1206≥0.47μF, 1210≥3.3μF, 1812≥4.7μF, 1825≥4.7μF, 2220≥4.7μF, 2225≥4.7μF	≤5%	0201≥0.01μF, 1210≥4.7μF	≤10%	0402≥0.1μF, 0603>0.1μF, 0805≥1μF, 1206≥2.2μF, 1210≥10μF	35V	≤3.5%	≤10%	0603≥1μF, 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V	≤3.5%	≤5%	0201≥0.01μF, 0805≥1μF, 1210≥10μF	≤7%	0603≥0.33μF, 1206≥4.7μF	≤10%	0201≥0.1μF, 0402≥0.10μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥6.8μF, 1210≥22μF	16V	≤3.5%	≤12.5%	0402≥0.47μF	≤5%	0201≥0.01μF, 0402≥0.033μF, 0603≥0.15μF, 0805≥0.68μF, 1206≥2.2μF, 1210≥4.7μF	≤10%	0201≥0.1μF(0201/X7R/X7S≥0.022μF), 0402≥0.22μF, 0603≥0.68μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥22μF	10V	≤5%	≤10%	0201≥0.012μF, 0402≥0.33μF(0402/X7R/X7S≥0.22μF), 0603≥0.33μF, 0805≥2.2μF, 1206≥2.2μF, 1210≥22μF				≤15%	0201≥0.1μF, 0402≥1μF
		Rated	D.F.≤	Exception of D.F.≤																																																
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		* Insulation Resistance. <table><tr><th>Rated Vol.(V)</th><th>Apply Voltage</th><th>Charge Time</th></tr><tr><td>≤100</td><td>1 times of U_R</td><td>Max. 120 sec.</td></tr><tr><td>200≤V≤500</td><td>1 times of U_R</td><td>60 sec.</td></tr><tr><td>>500</td><td>500Vdc</td><td>60 sec.</td></tr></table>	Rated Vol.(V)	Apply Voltage	Charge Time	≤100	1 times of U _R	Max. 120 sec.	200≤V≤500	1 times of U _R	60 sec.	>500	500Vdc	60 sec.	* I.R. : ≥10GΩ or RxC≥500Ω-F, whichever is smaller. Class II (X7R/X7S) <table><tr><th>Rated voltage</th><th>I.R.</th></tr><tr><td>100V : All X7R/X7S</td><td rowspan="6">≥10GΩ or RxC≥100Ω-F, whichever is smaller</td></tr><tr><td>50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF, 1812≥4.7μF, 1825≥4.7μF, 2220≥4.7μF, 2225≥4.7μF,</td></tr><tr><td>35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF</td></tr><tr><td>25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥1μF, 1210≥10μF</td></tr><tr><td>16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF</td></tr><tr><td>10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF</td></tr></table>	Rated voltage	I.R.	100V : All X7R/X7S	≥10GΩ or RxC≥100Ω-F, whichever is smaller	50V : 0402>0.01μF, 0603≥1μF, 0805≥1μF, 1206≥1.0μF, 1210≥4.7μF, 1812≥4.7μF, 1825≥4.7μF, 2220≥4.7μF, 2225≥4.7μF,	35V : 0805≥2.2μF, 1206≥2.2μF, 1210≥10μF	25V : 0402≥1μF, 0603≥2.2μF, 0805≥2.2μF, 1206≥1μF, 1210≥10μF	16V : 0402≥0.22μF, 0603≥1μF, 0805≥2.2μF, 1206≥10μF, 1210≥47μF	10V : 0402≥0.47μF, 0603≥0.47μF, 0805≥2.2μF, 1206≥4.7μF, 1210≥47μF																												
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		* Dielectric Strength. <table><tr><th>Rated Voltage</th><th>Condition</th></tr><tr><td>≤100</td><td>2.5 times of U_R</td></tr><tr><td>100<V≤250</td><td>2.0 times of U_R</td></tr><tr><td>250<V≤500</td><td>1.5 times of U_R</td></tr><tr><td>630≤V≤3000V</td><td>1.2 times of U_R</td></tr><tr><td>3000<V≤5000V</td><td>1.1 times of U_R</td></tr><tr><td>≥5000V</td><td>1.0 times of U_R</td></tr></table> * Duration 1~5 sec, charge and discharge current less than 50mA. * Test in insulating fluid for rated voltage ≥1KV products.	Rated Voltage	Condition	≤100	2.5 times of U _R	100<V≤250	2.0 times of U _R	250<V≤500	1.5 times of U _R	630≤V≤3000V	1.2 times of U _R	3000<V≤5000V	1.1 times of U _R	≥5000V	1.0 times of U _R	* No evidence of damage or flash over during test.																																			
Rated Voltage	Condition																																																			
≤100	2.5 times of U _R																																																			
100<V≤250	2.0 times of U _R																																																			
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≥5000V	1.0 times of U _R																																																			
		* Temperature Coefficient (With no electrical load) Operation temperature : -55~125°C at 25°C.	* Capacitance change : C0G : Within ±30ppm/°C. X7R within ±15%. X7S within ±22%.																																																	



8. RELIABILITY TEST CONDITIONS AND REQUIREMENTS

No.	AEC-Q200 Test Item	AEC-Q200 Test Condition	Requirements
17	Board Flex AEC-Q200-005	* The middle part of substrate shall be pressurized by means of the pressurizing rod at a rate of about 1mm per second until the deflection becomes 5mm for product size < 1808, 3mm for product size ≥ 1808 and then the pressure shall be maintained for 60±1 sec. * Measurement to be made after keeping at room temp. for 24±2 hrs.	* No remarkable damage. * Capacitance within the specified tolerance. * D.F. : ≤100% of initial requirement. * I.R. : ≥100% of initial requirement. (This capacitance change means the change of capacitance under specified flexure of substrate from the capacitance measured before the test)
18	Terminal Strength AEC-Q200-006	* Pressurizing force : 2N (0402), 5N(0603), 10N(0805), 17.7N(≥1206). * Test time : 60±1 sec.	* No remarkable damage or removal of the terminations. * Capacitance within the specified tolerance. * Q/D.F. value : To meet the initial requirement. * I.R. : To meet the initial requirement.
19	Beam Load Test AEC-Q200-003	* Break strength test. * Beam speed : 2.5±0.25 mm/sec.	* The chip endure following force : Chip length ≤2.5mm : Thickness >0.5mm (20N), ≤0.5mm (8N). Chip length ≥3.2mm : Thickness ≥1.25mm (54.5N), <1.25mm (15N).



9. PACKAGE DIMENSION AND QUANTITY

Size	Thickness (mm)	Paper tape		Plastic tape	
		7" reel	13" reel	7" reel	13" reel
0402 (1005)	0.50±0.20	10k	-	-	-
0603(1608)	0.50±0.10	4k	-	-	-
	0.80±0.07	4k	15k	-	-
	0.80 +0.15/-0.10	4k	15k	-	-
0805(2012)	0.50±0.10	4k	15k	-	-
	0.60±0.10	4k	15k	-	-
	0.80±0.10	4k	15k	-	-
	0.85±0.10	4k	15k	-	-
	1.25±0.10	-	-	3k	10k
	1.25±0.20	-	-	3k	10k
1206(3216)	0.80±0.10	4k	15k	-	-
	0.85±0.10	4k	15k	-	-
	0.95±0.10	-	-	3k	10k
	1.15±0.15	-	-	3k	10k
	1.25±0.10	-	-	3k	10k
	1.60±0.20	-	-	2k	9k
1210(3225)	1.60 +0.30/-0.10	-	-	2k	9k
	0.85±0.10	-	-	3k	10k
	0.95±0.10	-	-	3k	10k
	1.25±0.10	-	-	3k	10k
	1.60±0.20	-	-	2k	-
	2.00±0.20	-	-	1k	6k
1808(4520)	2.50±0.30	-	-	1k	6k
	1.25±0.10	-	-	2k	10k
	1.60±0.20	-	-	2k	8k
1812(4532)	2.00±0.20	-	-	1k	6k
	1.25±0.10	-	-	1k	5k
	1.60±0.20	-	-	1k	-
	2.00±0.20	-	-	1k	-
	2.50±0.30	-	-	0.5k	3k
	2.80±0.30	-	-	0.5k	-
2220(5750)	1.60±0.20	-	-	1k	-
	2.00±0.20	-	-	1k	-
	2.50±0.30	-	-	0.5k	-
	2.80±0.30	-	-	0.5k	-

Unit : pcs



9. PACKAGE DIMENSION AND QUANTITY

9.1. EMBOSSED TAPE DIMENSIONS

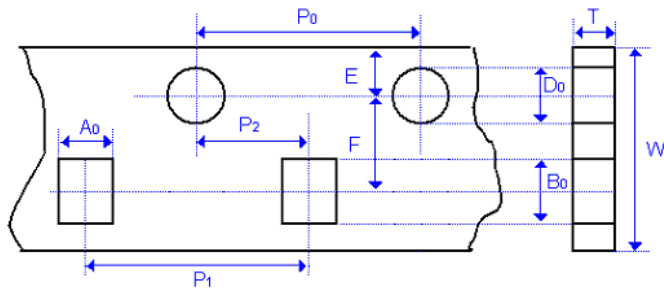


Fig. 9.1 The dimension of paper tape

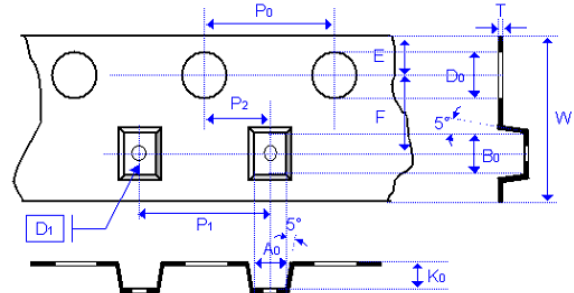


Fig. 9.2 The dimension of plastic tape

Size	0402	0603		0805	
Chip Thickness	0.50±0.20	0.80±0.07	0.80 +0.15/-0.1	0.80±0.10	1.25±0.10 1.25±0.20
A ₀	0.7+/-0.20	1.00 +0.05/-0.1	1.02 +0.05/-0.1	1.50±0.10	<1.65
B ₀	1.20±0.20	1.80±0.10	1.80±0.10	2.30±0.10	<2.40
T	≤0.80	0.95±0.05	0.97±0.05	0.95±0.05	0.23±0.05
K ₀	-	-	-	-	<2.50
W	8.00±0.30	8.00±0.10	8.00±0.10	8.00±0.10	8.00±0.10
P ₀	4.00±0.10	4.00±0.10	4.00±0.10	4.00±0.10	4.00±0.10
10xP ₀	40.00±0.10	40.00±0.20	40.00±0.20	40.00±0.20	40.00±0.20
P ₁	2.00±0.05	4.00±0.10	4.00±0.10	4.00±0.10	4.00±0.10
P ₂	2.00±0.05	2.00±0.05	2.00±0.05	2.00±0.05	2.00±0.05
D ₀	1.50+0.1/-0	1.55±0.05	1.55±0.05	1.55±0.05	1.50 +0.10/-0
D ₁	-	-	-	-	1.00±0.10
E	1.75±0.10	1.75±0.05	1.75±0.05	1.75±0.05	1.75±0.10
F	3.50±0.05	3.50±0.05	3.50±0.05	3.50±0.05	3.50±0.05
Unit :	mm	mm	mm	mm	mm

Size	1206			1210		1812	
Chip Thickness	0.80±0.10	0.95±0.10 1.25±0.10	1.60±0.20 1.60+0.3/-0.1	0.95±0.10 1.25±0.10 1.60±0.20 2.00±0.20	2.50±0.30	1.25±0.10 1.60±0.20 2.00±0.20	2.50±0.30 2.80±0.30
A ₀	2.00±0.10	<2.00	<2.00	<3.05	<3.10	<3.90	<3.90
B ₀	3.50±0.10	<3.60	<3.70	<3.80	<4.00	<5.30	<5.30
T	0.95±0.05	0.23±0.05	0.23±0.05	0.23±0.05	0.23±0.05	0.25±0.05	0.25±0.05
K ₀	-	<2.50	<2.50	<2.50	<3.50	<2.50	<3.00
W	8.00±0.10	8.00±0.10	8.00±0.10	8.00±0.10	8.00±0.10	12.00±0.20	12.00±0.20
P ₀	4.00±0.10	4.00±0.10	4.00±0.10	4.00±0.100	4.00±0.10	4.00±0.10	4.00±0.10
10xP ₀	40.00±0.20	40.00±0.20	40.00±0.20	40.00±0.20	40.00±0.20	40.00±0.20	40.00±0.20
P ₁	4.00±0.10	4.00±0.10	4.00±0.10	4.00±0.10	4.00±0.10	8.00±0.10	8.00±0.10
P ₂	2.00±0.05	2.00±0.05	2.00±0.05	2.00±0.05	2.00±0.05	2.00±0.05	2.00±0.05
D ₀	1.55±0.05	1.50 +0.10/-0	1.50 +0.10/-0	1.50 +0.10/-0	1.50 +0.10/-0	1.50 +0.10/-0	1.50 +0.10/-0
D ₁	-	1.00±0.10	1.00±0.10	1.00±0.10	1.00±0.10	1.50±0.10	1.50±0.10
E	1.75±0.10	1.75±0.10	1.75±0.10	1.75±0.10	1.75±0.10	1.75±0.10	1.75±0.10
F	3.50±0.05	3.50±0.05	3.50±0.05	3.50±0.05	3.50±0.05	5.50±0.05	5.50±0.05
Unit :	mm	mm	mm	mm	mm	mm	mm



9. PACKAGE DIMENSION AND QUANTITY

Size	2220	
Chip Thickness	1.40±0.15	2.50±0.30
	1.60±0.20	2.80±0.30
	2.00±0.20	
A ₀	<5.80	<5.80
B ₀	<6.50	<6.50
T	0.30±0.10	0.30±0.10
K ₀	<2.50	<3.10
W	12.00±0.20	12.00±0.20
P ₀	4.00±0.10	4.00±0.10
10xP ₀	40.00±0.20	40.00±0.20
P ₁	8.00±0.10	8.00±0.10
P ₂	2.00±0.05	2.00±0.05
D ₀	1.50 +0.10/-0	1.50 +0.10/-0
D ₁	1.50±0.10	1.50±0.10
E	1.75±0.10	1.75±0.10
F	5.50±0.05	5.50±0.05
Unit :	mm	mm

9.3. REEL DIMENSIONS

Size	0402, 0603, 0805, 1206, 1210		1808, 1812, 2220
Reel size	7"	13"	7"
C	13.0 +0.5/-0.2	13.0 +0.7/-0.3	13.0 +0.5/-0.2
W ₁	8.4 +1.5/-0	8.4 +2.0/-0	12.4 +2.0/-0
W	14.4max	14.4max	shall accommodate tape width without interference
A	178.0 ±0.10	330.0 ±1.0	178.0 ±0.10
N	60.0 +1.0/-0	100 ±1.0	60.0 +1.0/-0

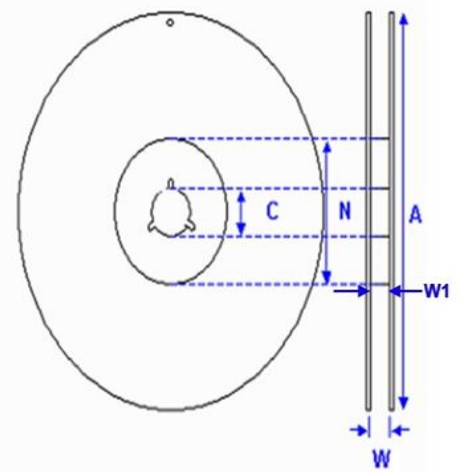


Fig. 9.3 The dimension of reel



10. APPLICATION NOTES

STORAGE

To prevent the damage of solderability of terminations, the following storage conditions are recommended :

Indoors under 5 ~ 40°C and 20% ~ 70% RH.

No harmful gases containing sulfuric acid, ammonia, hydrogen sulfide or chlorine.

Packaging should not be opened until the capacitors are required for use. If opened, the pack should be re-sealed as soon as is practicable. Taped product should be stored out of direct sunlight, which might promote deterioration in tape or adhesion performance. The product is recommended to be used within 12 months after shipment and checked the solderability before use.

HANDLING

Chip capacitors are dense, hard, brittle, and abrasive materials. They are liable to suffer mechanical damage, in the form of cracks or chips. Chip Capacitors should be handled with care to avoid contamination or damage. To use vacuum or plastic tweezers to pick up or plastic tweezers is recommended for manual placement. Tape and reeled packages are suitable for automatic pick and placement machine.

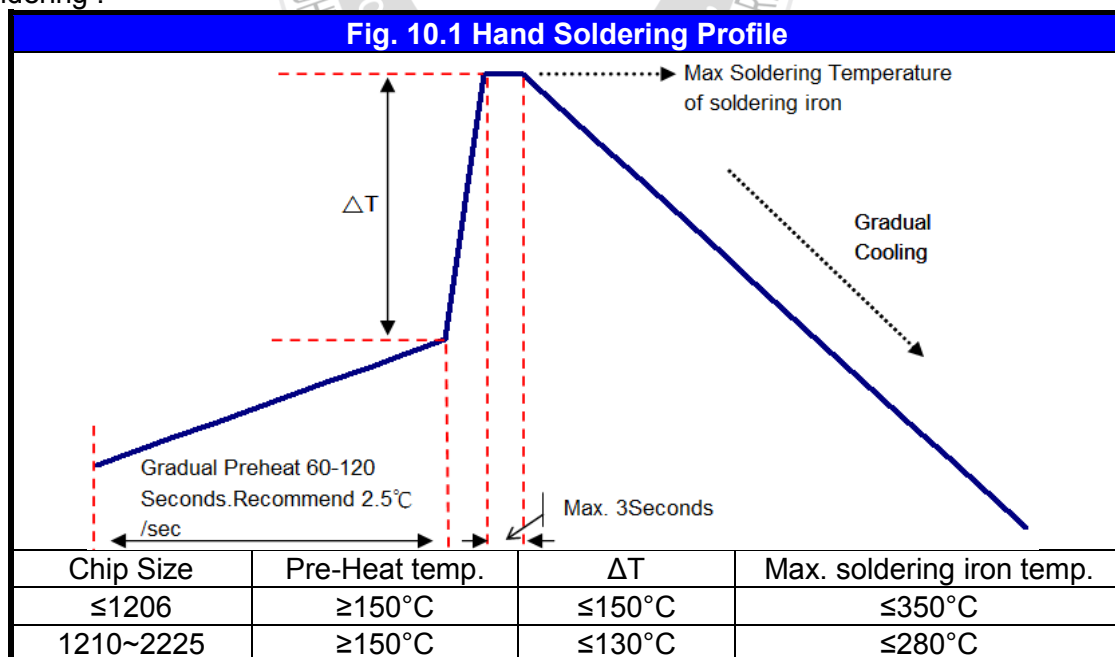
PREHEAT

In order to minimize the risk of thermal shock during soldering, a carefully controlled preheat is required. The rate of preheat should not exceed 3°C per second.

SOLDERING

Use mildly activated rosin fluxes do not use activated flux. The amount of solder in each solder joint should be controlled to prevent the damage of chip capacitors caused by the stress between solder, chips, and substrate.

a.) Hand soldering :



* Soldering iron tip diameter ≤1.0 mm and wattage max. 20W.

* The Capacitors shall be pre-heated and that the temperature gradient between the devices and the tip of the soldering iron.

* The required amount of solder shall be melted on the soldering tip.

* The tip of iron should not contact the ceramic body directly.

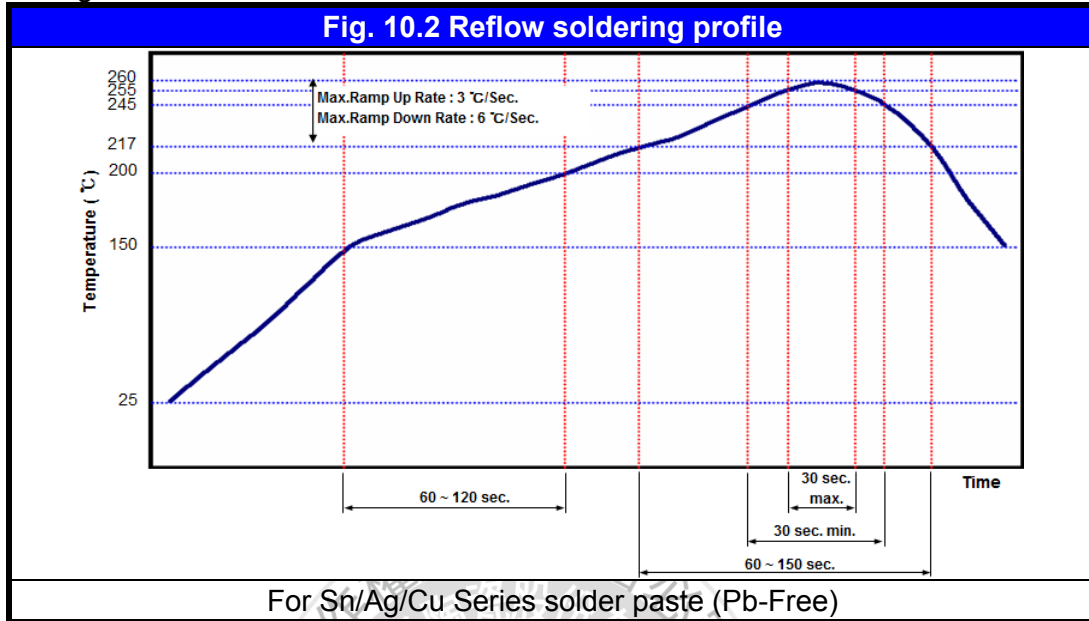
* The Capacitors shall be cooled gradually at room temperature after soldering.

* Forced air cooling is not allowed.

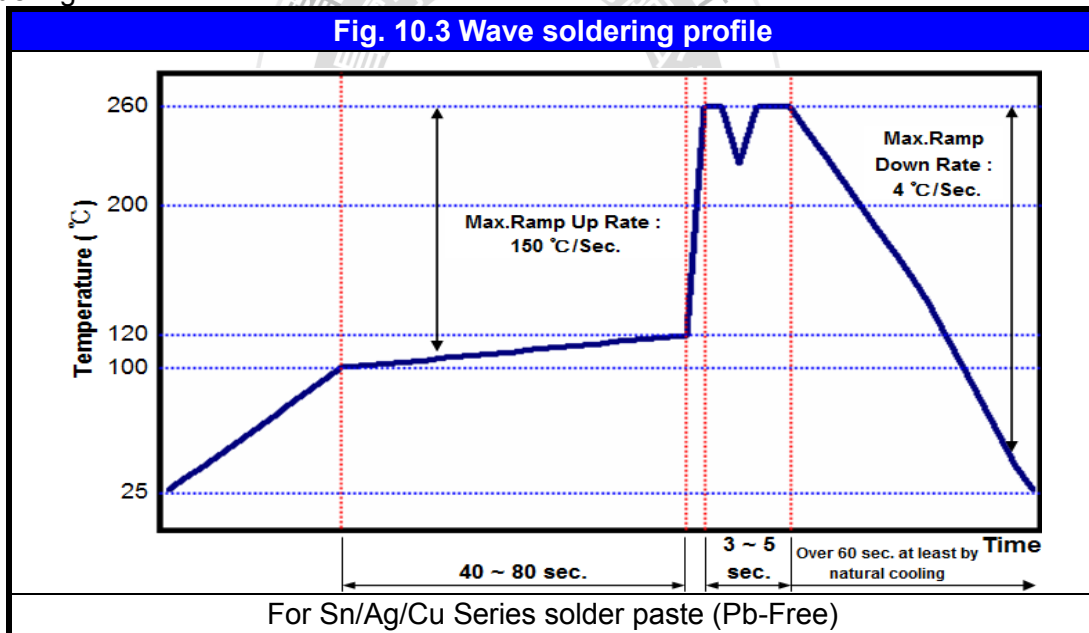


10. APPLICATION NOTES

b.) Reflow soldering :



c.) Wave soldering :



Soldering conditions :

Class I :

Size Inch (mm)	Temper. Cher.	Capacitance	Condition	
			Wave	Reflow
≤0402 (1005)	Class I	All	X	O
0603 (1608)	Class I	All	O	O
0805 (2012)	Class I	All	O	O
1206 (3216)	Class I	Thickness ≤ 0.95mm	O	O
		Thickness >0.95mm	X	O
≥1210 (3225)	Class I	All	X	O
Coating Products	All	All	X	O



10. APPLICATION NOTES

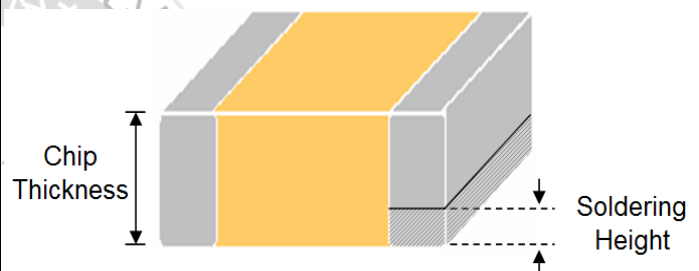
Soldering conditions :

Class II :

Size Inch (mm)	Temper. Cher.	Capacitance	Condition	
			Wave	Reflow
≤0402 (1005)	Class II	All	X	O
0603 (1608)	Class II	Cap. <2.2μF	O	O
		Cap. ≥2.2μF	X	O
0805 (2012)	Class II	Thickness ≤ 0.95mm	O	O
		Thickness > 0.95mm	X	O
1206 (3216)	Class II	Thickness ≤ 0.95mm	O	O
		Thickness > 0.95mm	X	O
≥1210 (3225)	Class II	All	X	O
Coating Products	All	All	X	O

Soldering height :

The solder climbing minimum height is suggesting to 25% of chip thickness or 500um whichever is less.
(Reference from IPC-610E)



COOLING

After soldering, cool the chips and the substrate gradually to room temperature. Natural cooling in air is recommended to minimize stress in the solder joint.

CLEANING

All flux residues must be removed by using suitable electronic-grade vapor-cleaning solvents to eliminate contamination that could cause electrolytic surface corrosion. Good results can be obtained by using ultrasonic cleaning of the solvent. The choice of the proper system is depends upon many factors such as component mix, flux, and solder paste and assembly method. The ability of the cleaning system to remove flux residues and contamination from under the chips is very important.

Notice of MT Series

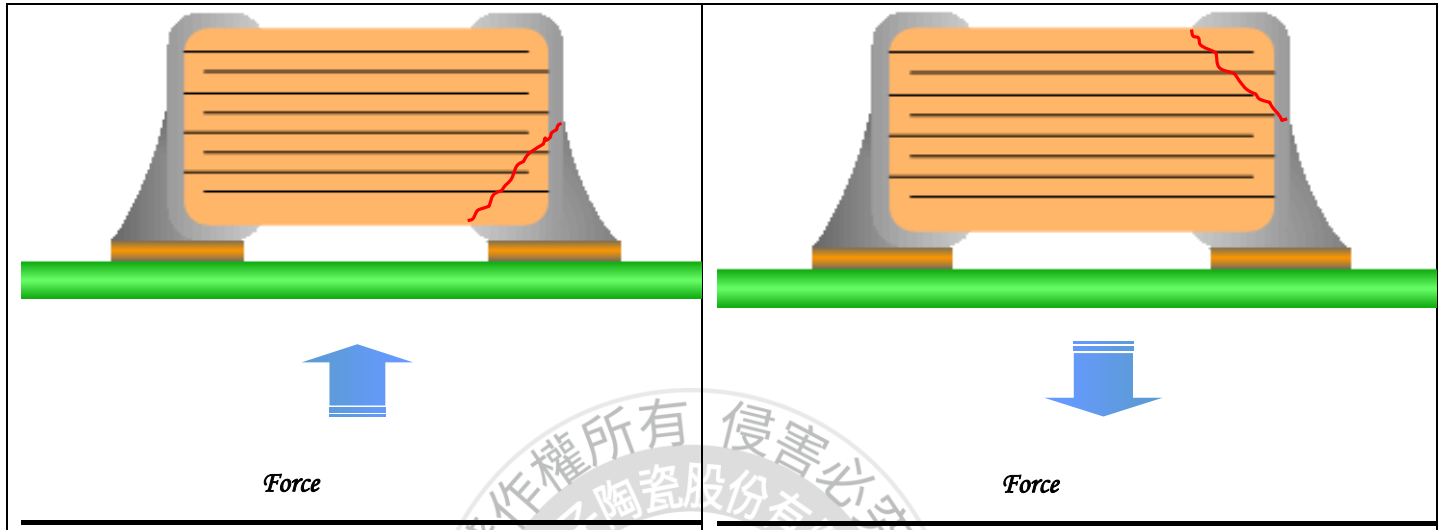
The standard AEC-Q200 series capacitors are mainly used on general automotive equipment without safety considerations. Please select SAFETY concern type or contact our company in advanced if you intend to use capacitor for designing the equipment which may damage itself and the safety of third party. If necessary, please consider to add the protect circuit in devising process and obtaining fully safety evaluation. The contents of the acknowledgments only used for our parent company, marketing subsidiaries and official marketing agents who purchase our products. Not applicable for the other nonofficial channels.

Surface coating products are not suitable cleaning/washing by solvent



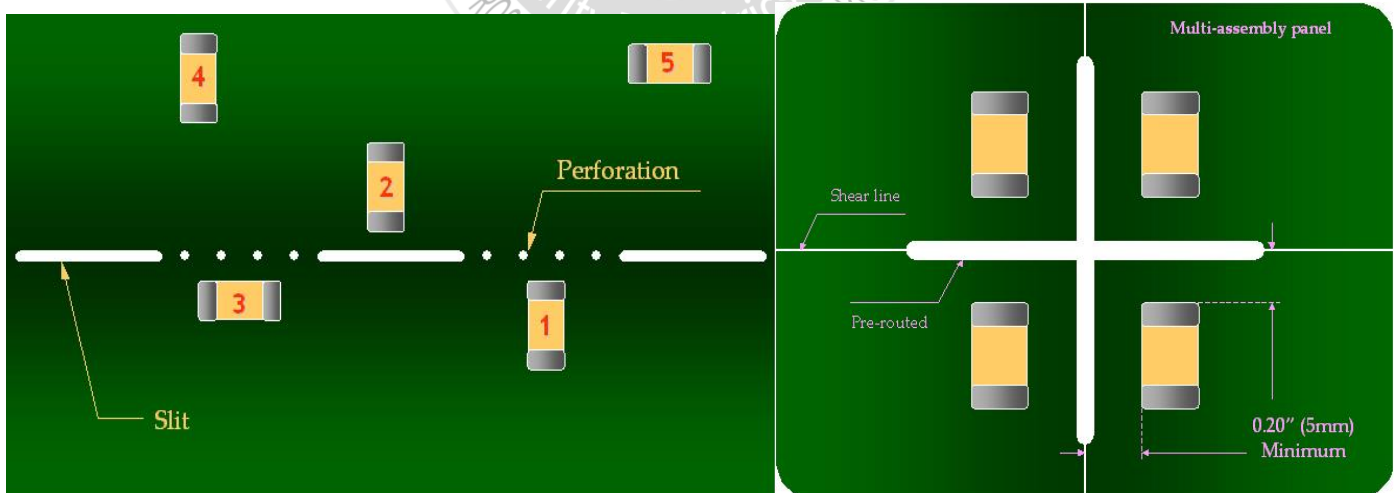
11. Typical Bending Cracks of MLCC

MLCC ceramic body is consisted of rigidity material. It will be suffered compressive and tensional stress when the carried board is bended. If the suffered stress is over ceramic body strength, the bending crack is occurred. **Therefore, the bending crack will be only occurred after soldering process.**



12. The stress v.s. position on PCB during bending

Chip mounting close to board separation point

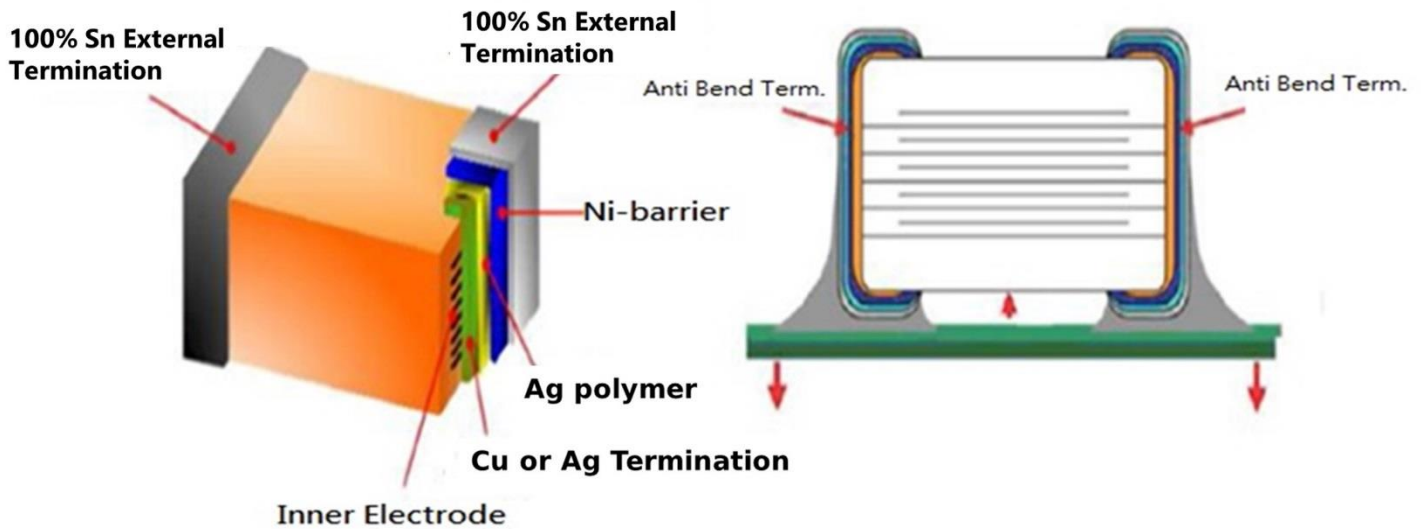


Magnitude of stress $1 > 2 \approx 3 > 4 > 5$



13. Structure

PDC soft termination series is added a special termination material(Ultra-Buffer or Anti-Bend)between ceramic body and Ni-barrier that can absorb mechanical stress to prevent bending crack occurred



14. Illustration of Bending Test

